Experiment: Installation and Configuration of Hadoop

Aim

To install and configure Apache Hadoop on a computer and understand its basic working.

Theory

Apache Hadoop is an open-source framework used to store and process large amounts of data across multiple computers. It is designed to handle Big Data efficiently by distributing data and processing tasks across a cluster of machines.

Hadoop follows a distributed computing approach, which means that data is divided into smaller parts and stored on different machines. It provides reliable and scalable data storage and processing.

Hadoop is mainly based on the HDFS (Hadoop Distributed File System) and MapReduce programming model. HDFS stores large datasets across multiple machines, while MapReduce processes the stored data in parallel.

Components of Hadoop

1. HDFS (Hadoop Distributed File System): Stores large files across multiple machines by dividing them into blocks. It provides reliable and distributed storage.
2. MapReduce: A programming model used to process large datasets in parallel. It works through two main phases: Map and Reduce.
3. YARN (Yet Another Resource Negotiator): Manages system resources and schedules jobs in the Hadoop cluster.
4. Hadoop Common: Provides common libraries and utilities required by other Hadoop components.

Installation and Configuration of Hadoop

The basic steps involved in installing Hadoop in a standalone or pseudo-distributed environment are:

1. Install Java Development Kit (JDK), as Hadoop requires Java to run.
2. Download the Apache Hadoop package from the official website: https://hadoop.apache.org/releases.html

3. Extract the downloaded Hadoop package into the required directory.
4. Set environment variables such as JAVA_HOME and HADOOP_HOME.
5. Configure Hadoop files such as core-site.xml, hdfs-site.xml, mapred-site.xml, and yarn-site.xml.
6. Configure SSH if required for running Hadoop services.
7. Format the NameNode using the following command:

hdfs namenode -format

8. Start HDFS and YARN services using:

start-dfs.sh

start-yarn.sh

9. Verify Hadoop installation using:

hadoop version

Basic Hadoop Commands

Some commonly used Hadoop commands are:

Command	Purpose
hadoop version	Displays Hadoop version
hdfs dfs -ls /	Lists files and directories in HDFS
hdfs dfs -mkdir /input	Creates a directory in HDFS
hdfs dfs -put file.txt /input	Uploads a file to HDFS
hdfs dfs -get /input/file.txt	Downloads a file from HDFS
hdfs dfs -cat /input/file.txt	Displays file contents
hdfs dfs -rm /input/file.txt	Deletes a file from HDFS

Advantages of Hadoop

- Handles large volumes of data efficiently.
- Provides distributed storage and parallel processing.
- Supports scalability by adding more machines.

- Offers fault tolerance through data replication.
- Supports structured, semi-structured, and unstructured data.

Conclusion

Apache Hadoop is a powerful framework used for storing and processing large datasets in a distributed environment. Its installation and configuration involve setting up Java, configuring Hadoop components, and starting the required services. Hadoop provides scalable, reliable, and cost-effective solutions for Big Data processing.